

2017-09-19 TRAFFIC ANALYSIS EXERCISE - ANSWERS

EXAMPLE OF AN INCIDENT WRITE-UP

On Tuesday 2017-09-19, a Windows computer was infected with Loki Bot malware after viewing an email from the user's personal webmail. Details follow:

- Date/Time: 2017-09-19 at 16:31 UTC
- User name/account name: Grace.Kelly
- Host name: Kelly-PC
- Host IP address: 192.168.1.14
- Host MAC address: 00:22:15:63:c9:5a (AsustekC_63:c9:5a)

The source of this infection was a message sent to the user's Yahoo email account. The email contains a malicious Microsoft Excel spreadsheet.

- Date/Time: 2017-09-19 at 10:45 UTC
- Subject: FW: Request for quotation - REMINDER
- Sending email address: "sales" <otimofeeva@rshn-tver.ru>
- Message-Id: <20170919104533.B5783455E3E@mail.rshn-tver.ru>
- Attachment name: 6068741003028.XLS

Indicators of compromise (IOCs):

SHA256 hash: [9b4e631b3a41b0a9feea26dedf363731eb851b77b20b6caebf9f5394281c111d](#)

- File name: 6068741003028.XLS
- File size: 94,208 bytes
- Description: Microsoft Excel spreadsheet with malicious macro

SHA256 hash: [b7dee12df4e82bbf11c73443f4371977caa38ada969256ac0f7973f2bf9e51ee](#)

- File location: C:\Users\Grace.Kelly\AppData\Roaming\8225BA\ABF6AB.exe
- File location: <https://u.cubeupload.com/1MDiPJ.jpg>
- File size: 498,640 bytes
- Description: Microsoft Excel spreadsheet with malicious macro

Macro in the Excel spreadsheet retrieving Loki Bot malware:

- 46.4.115.108 port 80 - ***i.cubeupload.com*** - GET /1MDiPJ.jpg
- 46.4.115.108 port 443 (HTTPS) - ***i.cubeupload.com*** - GET /1MDiPJ.jpg

Loki Bot post-infection traffic:

- 31.31.196.236 port 80 - ***obiltrate.ru*** - POST /doit1/fre.php

INTRODUCTION

First off, some introductory notes for people relatively new to analyzing network traffic...

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In order to analyze network traffic, especially the IPv4 traffic in these exercises, you must be familiar with how network traffic works. I originally learned through self-study for the Network+ certification. That was probably my single-most important building block for this type of analysis. My choice of study guide was [Todd Lammle's book](#), but if you have yet to acquire this knowledge, I suggest any of the [CompTIA sanctioned study guides](#).

But that's just the basics. Understanding malicious network traffic requires plenty of practice. People can easily mistake legitimate network traffic for malicious activity, because there are so many weird-looking domains and URLs associated with regular computer operations.

Pcaps for my normal blog posts on [malware-traffic-analysis.net](#) have other traffic stripped away, so you only find traffic related to the malicious activity. However, these traffic analysis exercises have more traffic, which better reflects real-life situations an analyst might run into.

In most cases, reviewing network traffic for suspicious activity is like finding a needle in a haystack. A great deal of network traffic is generated by any network-enabled device, whether it's an Android phone, a MacBook laptop, or a Windows desktop. That's why finding malicious activity is often done through investigating alerts.

These alerts can come from a variety of methods. My first jobs in cyber security involved reviewing alerts from Snort-based rules like those for [Snort](#) or [Suricata](#). By late 2013, I'd discovered a Linux distro called [Security Onion](#) that makes it easy to set up Snort or Suricata for live monitoring or playback of pcaps to reveal alerts.

However, it's easier for people to submit a pcap to [VirusTotal](#) or [PacketTotal](#) and see what Snort-based alerts the traffic might trigger. Just be aware that other people can review the pcaps you submit to these sites. If your pcaps contain any sensitive data, you should avoid using these resources. Fortunately, my traffic analysis exercises don't have sensitive data, so you can easily submit them to either site.

Besides those online resources, I primarily use Wireshark to investigate network traffic. As usual, I suggest you set up Wireshark according to the tutorials I've posted at: <http://malware-traffic-analysis.net/tutorials/index.html>.

IP, MAC ADDRESS, HOST NAME, AND USER NAME

It's easy enough to find the IP address, MAC address, host name, and user name (or account name) for this traffic. When people retrieve network traffic, they most often request it for a specific IP address. In this case, the IP address is 192.168.1.14. The associated MAC address is 00:22:15:63:c9:5a.

How can you find out the host name and tie these all together? Search for DHCP traffic. Since DHCP traffic uses UDP port 67 and 68, you can filter on that in Wireshark.

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The image below shows the filter expression **udp.port eq 67**, but you can also use **udp.port eq 68**. You can also use **==** instead of **eq** in the filter.

Filter: **udp.port eq 67** Expression... Clear Apply Save Filter Filter

Date/Time	Src	port	Dst	port	Info
2017-09-19 16:26:56	0.0.0.0	68	255.255.255.255	67	DHCP Request - Transaction ID 0x87368fd
2017-09-19 16:27:02	192.168.1.14	68	255.255.255.255	67	DHCP Inform - Transaction ID 0x8807e98f
2017-09-19 16:28:17	192.168.1.14	68	255.255.255.255	67	DHCP Inform - Transaction ID 0x27051daa
2017-09-19 16:28:20	192.168.1.14	68	255.255.255.255	67	DHCP Inform - Transaction ID 0xa132a0fc

Frame 1: 348 bytes on wire (2784 bits), 348 bytes captured (2784 bits)

Ethernet II, Src: AsustekC_63:c9:5a (00:22:15:63:c9:5a), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Internet Protocol Version 4, Src: 0.0.0.0 (0.0.0.0), Dst: 255.255.255.255 (255.255.255.255)

User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)

Bootstrap Protocol (Request)

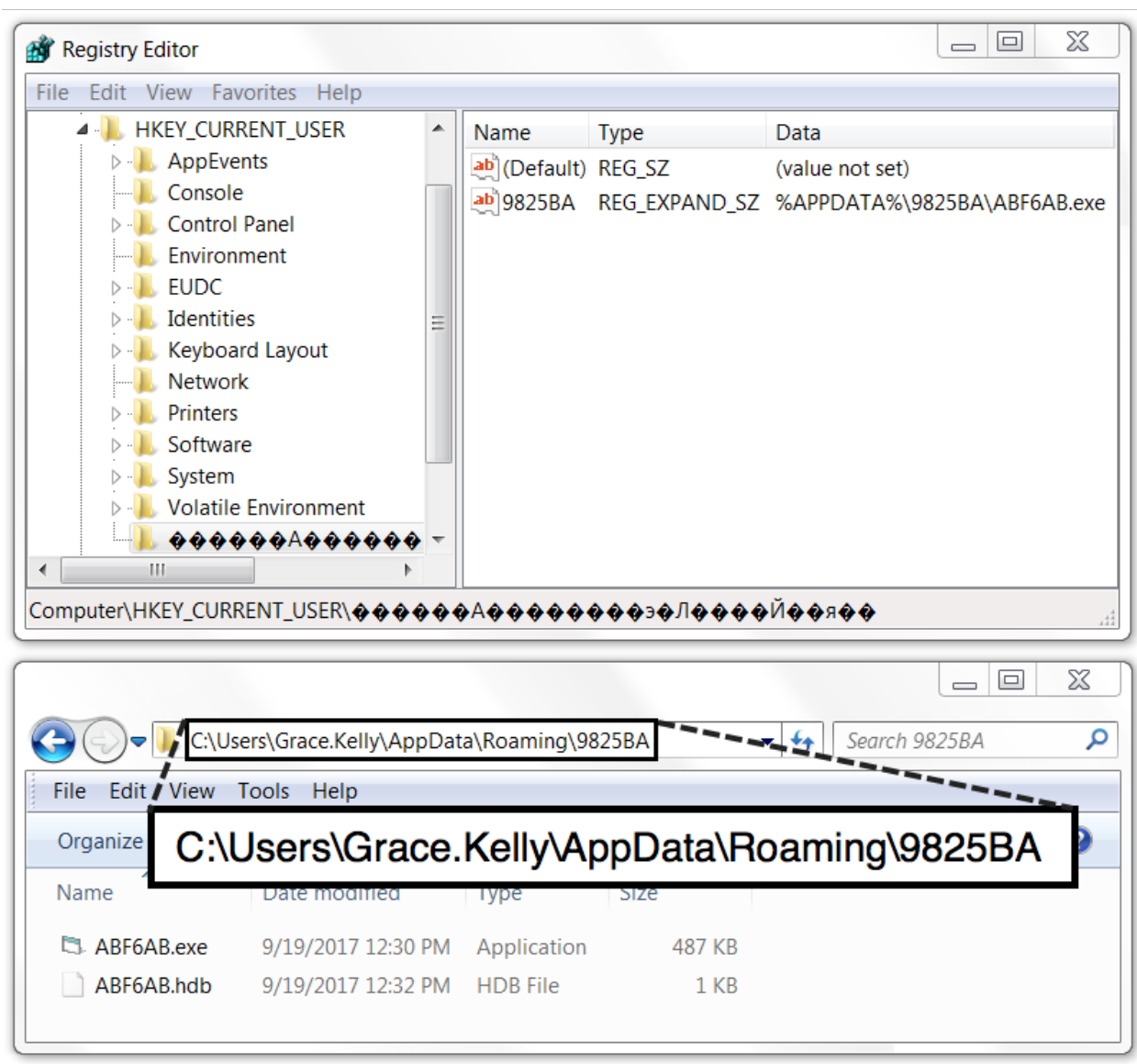
- Message type: Boot Request (1)
- Hardware type: Ethernet (0x01)
- Hardware address length: 6
- Hops: 0
- Transaction ID: 0x087368fd
- Seconds elapsed: 0
- Bootp flags: 0x8000 (Broadcast)
- Client IP address: 0.0.0.0 (0.0.0.0)
- Your (client) IP address: 0.0.0.0 (0.0.0.0)
- Next server IP address: 0.0.0.0 (0.0.0.0)
- Relay agent IP address: 0.0.0.0 (0.0.0.0)
- Client MAC address: AsustekC_63:c9:5a (00:22:15:63:c9:5a)
- Client hardware address padding: 00000000000000000000
- Server host name not given
- Boot file name not given
- Magic cookie: DHCP
- Option: (53) DHCP Message Type (Request)
- Option: (61) Client identifier
 - Length: 7
 - Hardware type: Ethernet (0x01)
 - Client MAC address: AsustekC_63:c9:5a (00:22:15:63:c9:5a)
- Option: (50) Requested IP Address
 - Length: 4
 - Requested IP Address: 192.168.1.14 (192.168.1.14)
- Option: (12) Host Name
 - Length: 8
 - Host Name: Kelly-PC

You can see where to expand the information and find the MAC address, IP address, and hostname. In this case, the pcap includes the host before it was assigned an IP, and the first frame shows 0.0.0.0 as its IP address. However, any of the following frames for DHCP should have the same data.

What about the user name (the Windows account name)? You can find that in the image Larry provided during the exercise scenario. The follow-up malware is stored in the computer's directory at **C:\Users\Grace.Kelly\AppData\Roaming\9825BA** that shows the user's account name as "Grace.Kelly". Also, the recipient's email address in

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the email Larry provided is **grace.kelly.2001@yahoo.com**. Furthermore, we'll also see the username when we review the post-infection traffic.



Shown above: The user's account name from the location for the malware.

```
Received: from [172.93.148.177] (unknown [172.93.148.177])
    by mail.rshn-tver.ru (Postfix) with ESMTPA id B5783455E3E
    for <grace.kelly.2001@yahoo.com>; Tue, 19 Sep 2017 13:45:33 +0300 (MSK)
Content-Type: multipart/mixed; boundary="====0192309506===="
MIME-Version: 1.0
Subject: FW: Request for quotation - REMINDER
To: grace.kelly.2001@yahoo.com
From: "sales" <otimofeeva@rshn-tver.ru>
Date: Tue, 19 Sep 2017 03:45:37 -0700
Message-Id: <20170919104533.B5783455E3E@mail.rshn-tver.ru>
```

Shown above: The user's Yahoo email address from the email Larry provided.

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MALWARE FROM THE EMAIL

I often review the pcap first, but we know from Larry that an email was involved. We also know it has an Excel spreadsheet as an attachment. You can find the spreadsheet by itself in the archive, or you can extract it from the email. If you submit the file to [VirusTotal](#), you'll find it has the following SHA256 file hash:

- 9b4e631b3a41b0a9feea26dedf363731eb851b77b20b6caebf9f5394281c111d

The spreadsheet has also been submitted to Payload Security's sandbox at [reverse.it](#). If you re-submit the file, it will go to the original analysis at:

- <https://www.reverse.it/sample/9b4e631b3a41b0a9feea26dedf363731eb851b77b20b6caebf9f5394281c111d?environmentId=100>

Look at the analysis, and you'll find the spreadsheet generates an HTTP GET request to 46.4.115.108 (i.cubeupload.com) for 1MDiPJ.jpg.

DNS Requests			
Domain	Address	Registrar	Country
u.cubeupload.com 	46.4.115.108	WEBFUSION LIMITED	Germany
i.cubeupload.com 	46.4.115.108	WEBFUSION LIMITED	Germany

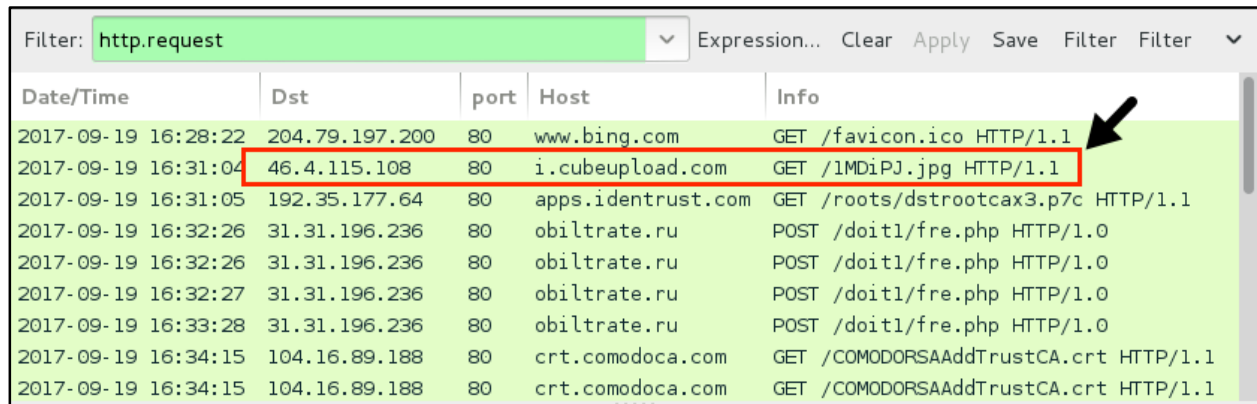
Contacted Hosts			
IP Address	Port/Protocol	Associated Process	Details
46.4.115.108 	80 TCP	powershell.exe PID: 3620	Germany
46.4.115.108 	443 TCP	powershell.exe PID: 3620	Germany

HTTP Traffic			
Endpoint	Request	URL	Data
46.4.115.108:80 (i.cubeupload.com)	 GET	/1MDiPJ.jpg	GET /1MDiPJ.jpg HTTP/1.1 Host: i.cubeupload.com Connection: Keep-Alive  301 Moved Permanently

Shown above: Traffic from the reverse.it analysis of the spreadsheet.

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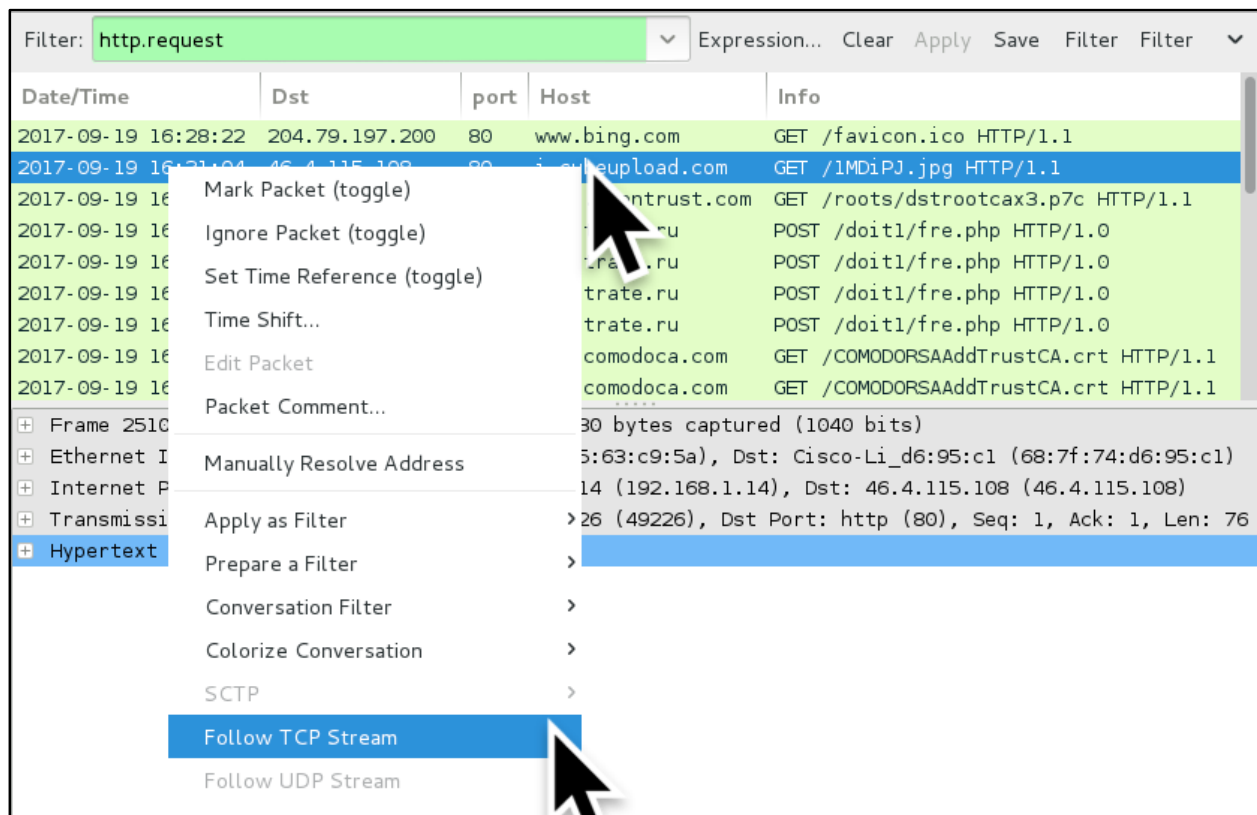
Let's look at the pcap now, so we can find the traffic. Filter on **http.request**, and we should find the same URL at the same IP address.



Date/Time	Dst	port	Host	Info
2017-09-19 16:28:22	204.79.197.200	80	www.bing.com	GET /favicon.ico HTTP/1.1
2017-09-19 16:31:04	46.4.115.108	80	i.cubeupload.com	GET /1MDiPJ.jpg HTTP/1.1
2017-09-19 16:31:05	192.35.177.64	80	apps.identrust.com	GET /roots/dstrootcax3.p7c HTTP/1.1
2017-09-19 16:32:26	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:32:26	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:32:27	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:33:28	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:34:15	104.16.89.188	80	crt.comodoca.com	GET /COMODORSAddTrustCA.crt HTTP/1.1
2017-09-19 16:34:15	104.16.89.188	80	crt.comodoca.com	GET /COMODORSAddTrustCA.crt HTTP/1.1

Shown above: Same URL from the reverse.it analysis is at the second HTTP request.

Right click to select that HTTP request, then left click to bring up the menu and follow the TCP stream. You can also follow the HTTP stream.



Date/Time	Dst	port	Host	Info
2017-09-19 16:28:22	204.79.197.200	80	www.bing.com	GET /favicon.ico HTTP/1.1
2017-09-19 16:31:04	46.4.115.108	80	i.cubeupload.com	GET /1MDiPJ.jpg HTTP/1.1
2017-09-19 16:31:05	192.35.177.64	80	apps.identrust.com	GET /roots/dstrootcax3.p7c HTTP/1.1
2017-09-19 16:32:26	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:32:26	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:32:27	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:33:28	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0
2017-09-19 16:34:15	104.16.89.188	80	crt.comodoca.com	GET /COMODORSAddTrustCA.crt HTTP/1.1
2017-09-19 16:34:15	104.16.89.188	80	crt.comodoca.com	GET /COMODORSAddTrustCA.crt HTTP/1.1

- Mark Packet (toggle)
- Ignore Packet (toggle)
- Set Time Reference (toggle)
- Time Shift...
- Edit Packet
- Packet Comment...
- Frame 2510
- Ethernet II
- Internet Protocol Version 4
- Transmission Control Protocol
- Hypertext Transfer Protocol
- Manually Resolve Address
- Apply as Filter
- Prepare a Filter
- Conversation Filter
- Colorize Conversation
- SCTP
- Follow TCP Stream
- Follow UDP Stream

Shown above: Following the TCP stream for that HTTP request.

You'll find a "301 Moved Permanently" response that shows the location at: **<https://u.cubeupload.com/1MDiPJ.jpg>**, so it's redirecting to the nearly same URL (at u.cubeupload.com instead of i.cubeupload.com), but as HTTPS over port 443.

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```
Stream Content
GET /lMDiPJ.jpg HTTP/1.1
Host: i.cubeupload.com
Connection: Keep-Alive

HTTP/1.1 301 Moved Permanently
Server: nginx/1.10.1 (Ubuntu)
Date: Tue, 19 Sep 2017 16:30:49 GMT
Content-Type: text/html
Content-Length: 194
Connection: keep-alive
Location: https://u.cubeupload.com/lMDiPJ.jpg

<html>
<head><title>301 Moved Permanently</title></head>
<body bgcolor="white">
<center><h1>301 Moved Permanently</h1></center>
<hr><center>nginx/1.10.1 (Ubuntu)</center>
</body>
</html>
```

Shown above: The HTTP response redirecting to an HTTPS URL.

The next TCP stream shows the HTTPS traffic, but it's encrypted, so we won't find out what it returned from the pcap alone. In the TCP stream, you'll find the string "Let's Encrypt" which is a free certificate service [often abused by malware distributors](#).

Filter: tcp.stream eq 67

Date/Time	Dst	port	Server Name	Info
2017-09-19 16:31:04	46.4.115.108	443	u.cubeupload.com	Client Hello

Follow TCP Stream (tcp.stream eq 67)

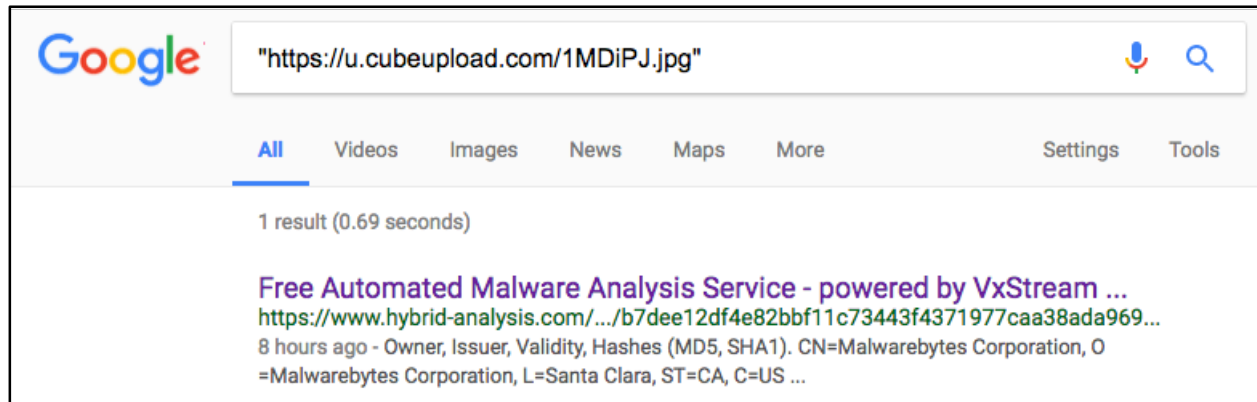
```
Stream Content
....s...o..Y.F:~W...Z.....Z.....Q.!...../.5...
.....
.2.8.....u.cubeupload.com.
.....]...Y...SQ...VE~.O._.....V~... \...n.. (.....'.N.....7.G.AB
$,..PEc.....-0..)0.....7~.tr...P0
..*.H..
....0J1.0...U....US1.0...U.
.
Let's Encrypt1#0!..U....Let's Encrypt Authority X30..
170731211300Z.
171029211300Z0.1.0...U....u.cubeupload.com0.."0
..*.H..
.....0..|
.....{.C.}>.....?4f.e..|
....Fzpj`P.`N.J.E....p7-O..&_..@....
$....K....wu....G.A..r..Z..Dh...>.a...A#m.....(.....[....f
%......e..X.O..V.R*f.8....P.F.(;..@..G..j[;...O.m.i.W....mr....=.....F.%
r.H1.r..S).l.j...s.'..
```

Shown above: HTTPS traffic using a Let's Encrypt certificate.

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Unfortunately, the reverse.it analysis doesn't show any further traffic after that initial HTTP request. However, we can search for "https://u.cubeupload.com/1MDiPJ.jpg" in Google--just remember to use the quotation marks, or you'll go directly to that URL.

The Google search shows another Payload Security analysis (this time at hybrid-analysis.com instead of reverse.it, but they're both the same thing).



Shown above: Search results on the HTTPS URL.

This analysis shows the HTTPS URL returns a Windows executable file for SHA256 hash b7dee12df4e82bbf11c73443f4371977caa38ada969256ac0f7973f2bf9e51ee. The executable file generates post-infection traffic to **obiltrate.ru**. That's the same SHA256 for the malware found on Kelly-PC at **C:\Users\Grace.Kelly\AppData\Roaming\8225BA\ABF6AB.exe**.

HTTP Traffic			
Endpoint	Request	URL	Data
31.31.196.236:80 (obiltrate.ru)	POST	/doit1/fre.php	POST /doit1/fre.php HTTP/1.0 User-Agent: Mozilla/4.08 (Charon; Inferno) Host: obiltrate.ru Accept: */* Content-Type: application/octet-stream Content-Encoding: binary Content-Key: 92AB4CD6 Content-Length: 192 Connection: close 404 Not Found More Details
31.31.196.236:80 (obiltrate.ru)	POST	/doit1/fre.php	POST /doit1/fre.php HTTP/1.0 User-Agent: Mozilla/4.08 (Charon; Inferno) Host: obiltrate.ru Accept: */* Content-Type: application/octet-stream Content-Encoding: binary Content-Key: 92AB4CD6 Content-Length: 192 Connection: close 404 Not Found More Details

Shown above: HTTP post-infection traffic from a Windows executable downloaded from https://u.cubeupload.com/1MDiPJ.jpg

We can easily find the same post-infection traffic in our pcap by filtering on **http.request** and giving the results a quick glance.

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Filter:	http.request	▼	Expression...	Clear	Apply	Save	Filter	▼
Date/Time	Dst	port	Host	Info				
2017-09-19 16:28:22	204.79.197.200	80	www.bing.com	GET /favicon.ico HTTP/1.1				
2017-09-19 16:31:04	46.4.115.108	80	i.cubeupload.com	GET /1MDiPJ.jpg HTTP/1.1				
2017-09-19 16:31:05	192.35.177.64	80	apps.identrust.com	GET /roots/dstrootcax3.p7c HTTP/1.1				
2017-09-19 16:32:26	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:32:26	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:32:27	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:33:28	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:34:15	104.16.89.188	80	crt.comodoca.com	GET /COMODORSAAddTrustCA.crt HTTP/1.1				
2017-09-19 16:34:15	104.16.89.188	80	crt.comodoca.com	GET /COMODORSAAddTrustCA.crt HTTP/1.1				
2017-09-19 16:34:15	23.64.151.213	80	ssl.trustwave.com	GET /issuers/STCA.crt HTTP/1.1				
2017-09-19 16:34:15	23.64.151.213	80	ssl.trustwave.com	GET /issuers/STCA.crt HTTP/1.1				
2017-09-19 16:34:15	162.208.20.184	80	geo-um.btrll.com	GET /v1/map/68e4318f0f7098c3/209157352				
2017-09-19 16:34:16	54.192.6.254	80	cache.btrll.com	GET /default/Pix-1x1.gif HTTP/1.1				
2017-09-19 16:34:16	162.208.20.184	80	geo-um.btrll.com	GET /v1/map/897cac40a52e7ede/FsbKeps21				
2017-09-19 16:34:28	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:35:29	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:36:30	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:37:30	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:38:31	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:39:31	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:40:32	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:41:33	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				
2017-09-19 16:42:33	31.31.196.236	80	obiltrate.ru	POST /doit1/fre.php HTTP/1.0				

Shown above: HTTP requests in exercise pcap show the same post-infection traffic.

How do we know what the malware is? Look further down in the hybrid-analysis.com results, and you'll find Emerging Threats alerts for Loki Bot.

Emerging Threats			
(TCP)		webresource or standalone version	
31.31.196.236:80 (TCP)	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	2021641
31.31.196.236:80 (TCP)	Hidden Category	Additional ETPro rules are available in the private webresource or standalone version	Hidden SID
31.31.196.236:80 (TCP)	Hidden Category	Additional ETPro rules are available in the private webresource or standalone version	Hidden SID
31.31.196.236:80 (TCP)	Hidden Category	Additional ETPro rules are available in the private webresource or standalone version	Hidden SID
31.31.196.236:80 (TCP)	A Network Trojan was detected	ET TROJAN Loki Bot Request for C2 Commands Detected M1	2024313
31.31.196.236:80	A Network Trojan was detected	ET TROJAN Loki Bot Request for C2 Commands Detected	2024318

ET rules applied using Suricata. ETPro rule matches (11 total) are hidden and available in the [private webresource](#) or standalone version.

Shown above: Emerging Threats alerts for Loki Bot on same IP as post-infection traffic.

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OTHER ONLINE TOOLS

As I mentioned in the introduction, you can submit pcaps to [VirusTotal](#) or [PacketTotal](#) and see what alerts the traffic might trigger. Some VirusTotal results follow:

The screenshot shows the VirusTotal interface for a file analysis. At the top, it says "No engines detected this file". Below this, file details are listed: SHA-256 hash, file name (2017-09-19-traffic-analysis-exercise.pcap), file size (5.38 MB), and last analysis time (2017-09-19 17:30:17 UTC). A green badge indicates "0/58" engines. The "Detection" tab is active, showing a table of engine results. The table has two columns for engine names and their detection status. Snort and Suricata show alerts, while all other engines show "Clean".

Detection	Details	Community	
Snort	7 alerts	Suricata	11 alerts
Ad-Aware	Clean	AegisLab	Clean
AhnLab-V3	Clean	ALYac	Clean
Antiy-AVL	Clean	Arcabit	Clean
Avast	Clean	Avast Mobile Security	Clean
AVC	Clean	Avira	Clean

Shown above: The "Detections" tab shows 7 Snort alerts and 11 Suricata alerts.

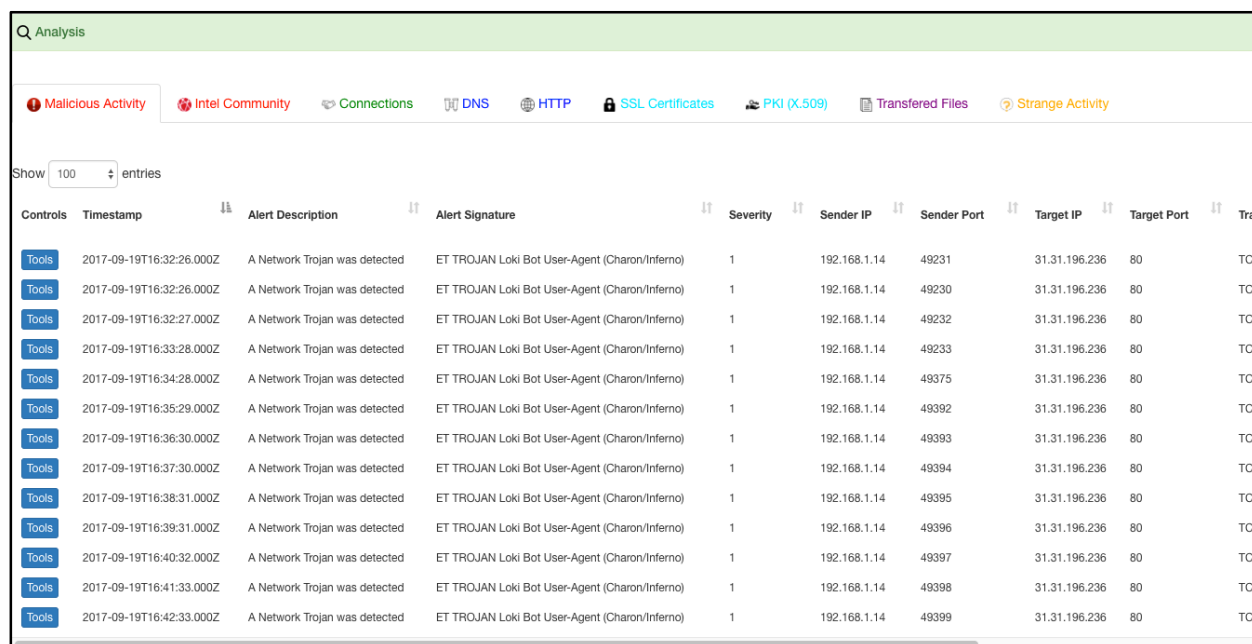
The screenshot shows the "Details" tab of the VirusTotal interface, displaying a list of alerts. The alerts are categorized into "Snort Alerts" and "Suricata Alerts". Under "Snort Alerts", there are five items: "Potentially Bad Traffic", "Attempted User Privilege Gain", "Potential Corporate Privacy Violation", "A Network Trojan was detected" (with a sub-item "BLACKLIST User-Agent known malicious user-agent string - Win.Trojan.Fareit [40066]"), and "Misc Attack". Under "Suricata Alerts", there are three items: "Potentially Bad Traffic", "A Network Trojan was Detected" (with a list of sub-items including "ET TROJAN Loki Bot User-Agent (Charon/Inferno) [2021641]", "ET TROJAN Loki Bot Application/Credential Data Exfiltration Detected M1 [2024312]", "ET TROJAN Loki Bot Request for C2 Commands Detected M1 [2024313]", "ET TROJAN Loki Bot Application/Credential Data Exfiltration Detected M2 [2024317]", "ET TROJAN Loki Bot Request for C2 Commands Detected M2 [2024318]", "ETPRO TROJAN Loki Bot Checkin [2815070]", and "ETPRO TROJAN Loki Bot Checkin M2 [2825766]"), and "Potential Corporate Privacy Violation". At the bottom, there is a "Misc activity" item.

Shown above: In the "Details" tab, scroll to the bottom to see Snort and Suricata alerts.

2017-09-19 TRAFFIC ANALYSIS EXERCISE - ANSWERS

VirusTotal uses both the Snort subscriber ruleset and the EmergingThreats Pro ruleset. On the other hand, PacketTotal uses only the EmergingThreats Open ruleset. It has a better reporting format than VirusTotal, though. After all, PacketTotal was designed specifically for packet analysis.

PacketTotal's analysis of the pcap can be found [here](#). The layout is great. PacketTotal correlates each alert to a specific TCP stream and HTTP request.



The screenshot shows the PacketTotal Analysis interface. At the top, there's a search bar and a navigation menu with tabs: Malicious Activity, Intel Community, Connections, DNS, HTTP, SSL Certificates, PKI (X.509), Transferred Files, and Strange Activity. Below the menu, there's a 'Show' dropdown set to 100 entries. The main table displays a list of alerts. Each row includes a 'Tools' button, a timestamp, an alert description, an alert signature, severity, sender IP, sender port, target IP, target port, and transport protocol.

Controls	Timestamp	Alert Description	Alert Signature	Severity	Sender IP	Sender Port	Target IP	Target Port	Tran
Tools	2017-09-19T16:32:26.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49231	31.31.196.236	80	TCP
Tools	2017-09-19T16:32:26.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49230	31.31.196.236	80	TCP
Tools	2017-09-19T16:32:27.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49232	31.31.196.236	80	TCP
Tools	2017-09-19T16:33:28.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49233	31.31.196.236	80	TCP
Tools	2017-09-19T16:34:28.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49375	31.31.196.236	80	TCP
Tools	2017-09-19T16:35:29.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49392	31.31.196.236	80	TCP
Tools	2017-09-19T16:36:30.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49393	31.31.196.236	80	TCP
Tools	2017-09-19T16:37:30.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49394	31.31.196.236	80	TCP
Tools	2017-09-19T16:38:31.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49395	31.31.196.236	80	TCP
Tools	2017-09-19T16:39:31.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49396	31.31.196.236	80	TCP
Tools	2017-09-19T16:40:32.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49397	31.31.196.236	80	TCP
Tools	2017-09-19T16:41:33.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49398	31.31.196.236	80	TCP
Tools	2017-09-19T16:42:33.000Z	A Network Trojan was detected	ET TROJAN Loki Bot User-Agent (Charon/Inferno)	1	192.168.1.14	49399	31.31.196.236	80	TCP

Shown above: Partial screenshot for PacketTotal's analysis of the exercise pcap.

FINAL WORDS

There's more you could examine in the pcap. For example, I didn't review any of the HTTPS activity before the infection, which shows traffic to **mail.yahoo.com**, indicating the victim checked Yahoo mail before the host was infected. But hopefully this document should answer any questions you might have about the infection traffic.

If you became frustrated as you went through this exercise, that's just part of the process. We won't learn unless we experience some sort failure and frustration. So if you're relatively new and found this hard, please don't get too discouraged. I always encourage people to keep at it.

Until next time...

- Brad Duncan